1. Encoding, Hashing, and Encryption

- What is the purpose of encoding? To convert data into a different format for storage or transmission. Encoding is not encryption and does not provide security by itself.

- How is hashing different from encryption? Hashing produces a fixed-length digest, while encryption transforms data into a form that can be reversed using a key.

- Which technique is designed to be one-way? Hashing.

- Which technique requires a key to recover the original data? Encryption.

2. Hash Algorithms

- What is a hash algorithm? A mathematical function that converts data of any size into a fixed-length hash value.

- Why are MD5 and SHA-1 no longer recommended for most security applications? Because they are vulnerable to collision attacks, making them unsuitable for modern security-sensitive uses.

- What is the SHA-2 family? A family of secure hash algorithms, including SHA-256 and SHA-512, widely used in cybersecurity.

- What is a hash collision? When two different inputs produce the same hash value.

3. Real-World Applications

- Why are passwords stored as hashes instead of plain text? To prevent attackers from directly reading passwords if a database is compromised. Passwords should be stored using secure, salted password-hashing algorithms such as Argon2 or bcrypt.

- How can hashes verify a downloaded file's integrity? Compare the file's calculated hash with the trusted hash provided by the source. If they match, the file's integrity is supported.

- What role do hashes play in digital signatures? They create a digest of the data that is signed using a private key, allowing the recipient to verify integrity and authenticity.

- Name three places where hashing is used in everyday computing.

1. Password storage.
2. Downloaded file integrity checks.
3. Digital signatures.